

A square-root bound for global binomial log-concavity

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Abstract

Let $\mathcal{F} \subseteq 2^{[N]}$ contain the empty set, and let M be its number of inclusion-maximal members. We prove that every integer $m \geq 4(N+1)^4 \sqrt{M}$ makes all coefficients of $(1+x)^m \sum_{S \in \mathcal{F}} x^{|S|}$ log-concave. No closure assumption on $\mathcal{F}$ is needed. The exponent $1/2$ of M cannot be reduced, even for independence complexes of trees, if arbitrary fixed polynomial factors in N are allowed. Consequently the worst global repair thresholds are $2^{N/4+o(N)}$ for forests, $3^{N/6+o(N)}$ for graphs, and $2^{N/2+o(N)}$ for simplicial complexes or arbitrary set families. The proof separates a coefficient-independent interior estimate from uniform endpoint estimates. All constants in the upper bound are explicit.

1 Statement and scope

For a nonzero polynomial P with nonnegative coefficients and positive constant coefficient, put

$$\sigma(P) = \min\{m \in \mathbb{Z}_{\geq 0} : (1+x)^m P(x) \text{ has a log-concave coefficient sequence}\}.$$

Log-concavity here includes the absence of internal zeros, as well as $c_k^2 \geq c_{k-1}c_{k+1}$. The assertions below prove finiteness in the cases under consideration. For $\mathcal{F} \subseteq 2^{[N]}$ containing $\emptyset$, write $P_{\mathcal{F}}(x) = \sum_{S \in \mathcal{F}} x^{|S|}$, and let $M(\mathcal{F})$ count its inclusion-maximal members.

Theorem 1 (Global square-root bound). *For $N \geq 1$ and every such family $\mathcal{F}$,*

$$\sigma(P_{\mathcal{F}}) \leq \left\lceil 4(N+1)^4 \sqrt{M(\mathcal{F})} \right\rceil.$$

In fact every integer at or above the displayed real bound makes the entire coefficient sequence strictly log-concave at its interior indices.

For a graph G , use its family of independent sets. Then $M(\mathcal{F}) = \text{mis}(G)$ and multiplying by $(1+x)^m$ adds m isolated vertices. This theorem repairs *all* log-concavity inequalities, not just the last one. For a simplicial complex the same operation is joining with a simplex on m new vertices, or equivalently taking m successive cones.

Position relative to existing work. Eventual log-concavity after multiplication by a binomial power is classical: Handelman [1, 2] treats a much wider existence question and determines the exact threshold $k^2 - 3$ for $(1+x)^m(1+x^k)$. The contribution derived here is an explicit bound governed by the number of maximal members, with matching exponential rates for several classes. The lower-bound tree construction is known from Galvin [3], and the maximal-independent-set estimates are classical [4]. A literature search has not established publication priority for the present quantitative statement. This is a self-contained research draft, not an independently reviewed or formally verified result.

2 The coefficient-independent interior

Write $B_r = \binom{m}{r}$, with $B_r = 0$ outside $0 \leq r \leq m$.

Lemma 2 (Pairwise binomial curvature). *Let $n \geq 1$, $L = n^2$, and $m \geq 2L$. If $L \leq r, s \leq m - L$ and $|r - s| \leq n$, then*

$$2B_r B_s > B_{r-1} B_{s+1} + B_{s-1} B_{r+1}.$$

Consequently, for any $P(x) = \sum_{i=0}^d a_i x^i$ with $a_i \geq 0$, $a_0 > 0$, and $d \leq n$, the coefficients $c_k = \sum_i a_i B_{k-i}$ satisfy $c_k^2 > c_{k-1} c_{k+1}$ whenever

$$n^2 + n < k < m + d - (n^2 + n).$$

Proof. Set $u_r = r/(m - r + 1)$ and $v_r = (m - r)/(r + 1)$. Direct subtraction gives

$$2 - u_r v_s - u_s v_r = \frac{(m+1)\{(r+1)(m-r+1) + (s+1)(m-s+1) - (m+1)(r-s)^2\}}{(r+1)(s+1)(m-r+1)(m-s+1)}.$$

The braces are strictly positive: their first two terms sum to at least $2(L+1)(m-L+1) \geq (L+1)(m+2) > L(m+1)$, while the subtracted term is at most $L(m+1)$. Multiply by $B_r B_s > 0$.

For the consequence, every $r = k - i$ belongs to $[L, m - L]$ and any two such values differ by at most $d \leq n$. Symmetrize the identity

$$c_k^2 - c_{k-1} c_{k+1} = \frac{1}{2} \sum_{i,j} a_i a_j \{2B_{k-i} B_{k-j} - B_{k-1-i} B_{k+1-j} - B_{k-1-j} B_{k+1-i}\}.$$

Every summand is nonnegative, and the $(0, 0)$ summand is positive. $\square$

3 Uniform control at both endpoints

Lemma 3 (Two-endpoint smoothing). *Let $n \geq 1$, $d \leq n$, $M \geq 1$, and $P(x) = \sum_{i=0}^d a_i x^i$ with $a_i \geq 0$ and $a_0 a_d > 0$. Suppose, for $2 \leq i \leq d$, that*

$$\frac{a_i}{a_0} \leq M(n+1)^i, \quad \frac{a_{d-i}}{a_d} \leq M(n+1)^i.$$

Then every integer $m \geq 4(n+1)^4 \sqrt{M}$ makes $(1+x)^m P$ strictly log-concave at every interior index.

Proof. If $d = 0$, this is the binomial sequence. Assume $d \geq 1$, and put $K = n(n+1)$. On the left, split

$$c_j = b_j + e_j, \quad b_j = a_0 B_j + a_1 B_{j-1}, \quad e_j = \sum_{i=2}^d a_i B_{j-i}.$$

For $0 \leq j \leq K+1$ and $i \leq j$,

$$\frac{B_{j-i}}{B_j} = \prod_{\ell=0}^{i-1} \frac{j-\ell}{m-j+\ell+1} \leq \left(\frac{K+1}{m-K} \right)^i.$$

Terms with $i > j$ vanish. Set $z = (n+1)(K+1)/(m-K)$. As $K \leq (n+1)^4 \sqrt{M}$ and $K+1 \leq (n+1)^2$,

$$z \leq \frac{1}{3(n+1)\sqrt{M}} \leq \frac{1}{6}, \quad 0 \leq \frac{e_j}{b_j} \leq \frac{e_j}{a_0 B_j} \leq M \sum_{i=2}^d z^i \leq 2Mz^2 \leq \frac{2}{9(n+1)^2} \leq \frac{1}{4K}. \quad (1)$$

All denominators are positive; the bound on m also ensures $K + 1 < m$.

The baseline satisfies

$$b_k^2 \geq \frac{k+1}{k} b_{k-1} b_{k+1} \quad (1 \leq k \leq K). \quad (2)$$

For completeness, write $u_j = j!b_j$. For $j \geq 1$,

$$u_j = (m)_{j-1} L_j, \quad L_j = (m - j + 1)a_0 + j a_1,$$

where $(m)_r$ is a falling factorial. The L_j are positive and affine, so $L_j^2 - L_{j-1} L_{j+1} = (a_1 - a_0)^2 \geq 0$. The falling-factorial factors are log-concave as well. This proves $u_j^2 \geq u_{j-1} u_{j+1}$ for $j \geq 2$ in the required range. At $j = 1$, direct calculation gives $u_1^2 - u_0 u_2 = m a_0^2 + a_1^2 \geq 0$. This is precisely (2).

With $\varepsilon = 1/(4K)$, (1)–(2) imply, for $1 \leq k \leq K$,

$$c_k^2 \geq b_k^2 \geq (1 + 1/K) b_{k-1} b_{k+1} > (1 + \varepsilon)^2 b_{k-1} b_{k+1} \geq c_{k-1} c_{k+1},$$

since $1 + 1/K > (1 + 1/(4K))^2$. Apply the identical argument to $x^d P(1/x)$ for the last K indices of the product, whose degree is $m + d$. All remaining indices satisfy Lemma 2. Positivity of all coefficients follows from $a_0 a_d > 0$ and $m \geq d$: the contributions of a_0 and a_d cover $[0, m]$ and $[d, m + d]$, respectively. $\square$

4 From maximal members to all coefficients

Proof of Theorem 1. Put $d = \max_{S \in \mathcal{F}} |S|$ and $P_{\mathcal{F}} = \sum a_i x^i$. We have $a_0 = 1$, $a_d \geq 1$, and $a_i \leq \binom{N}{i} \leq (N + 1)^i$. Every member of size $d - i$ extends to an inclusion-maximal member. The extension adds at most i elements. Each maximal member contains at most $\sum_{j=0}^i \binom{N}{j}$ subsets obtained by deleting at most i elements. Therefore

$$a_{d-i} \leq M \sum_{j=0}^i \binom{N}{j} \leq M(N + 1)^i. \quad (3)$$

The last inequality follows by encoding a subset of size at most i as its increasing list, padded with zeros to length i , over an alphabet of $N + 1$ symbols. This encoding is injective. Division by $a_d \geq 1$ only improves the estimate. Lemma 3, with $n = N$, now applies. The case $d = 0$ is already included. $\square$

Corollary 4 (A polynomial bound for few maximal members). *If $M(\mathcal{F}) \leq N^c$ for a fixed c , then $\sigma(P_{\mathcal{F}}) \leq \lceil 4(N + 1)^4 N^{c/2} \rceil$. This applies without assuming that the family is a simplicial complex.*

5 Sharp exponential rates

Let $S_{\text{for}}(N)$ and $S_{\text{gr}}(N)$ be the maximum of $\sigma(I(G; x))$ over forests and all graphs, respectively, of order at most N . Let $S_{\text{cx}}(N)$ and $S_{\text{fam}}(N)$ be the analogous maxima over simplicial complexes and set families containing the empty set, on at most N labelled elements.

Theorem 5. *As $N \rightarrow \infty$,*

$$\boxed{\begin{aligned} S_{\text{for}}(N) &= 2^{N/4+o(N)}, \\ S_{\text{gr}}(N) &= 3^{N/6+o(N)}, \\ S_{\text{cx}}(N) &= 2^{N/2+o(N)}, \\ S_{\text{fam}}(N) &= 2^{N/2+o(N)}. \end{aligned}}$$

These are thresholds for global log-concavity.

Proof. We give both sides.

Upper bounds. The classical bounds $\text{mis}(G) \leq 2^{N/2}$ for forests and $\text{mis}(G) \leq 3^{N/3}$ for graphs give

$$S_{\text{for}}(N) \leq \lceil 4(N+1)^4 2^{N/4} \rceil, \quad S_{\text{gr}}(N) \leq \lceil 4(N+1)^4 3^{N/6} \rceil. \quad (4)$$

Here are short proofs of the counting inputs. In a forest, an isolated vertex is forced into every maximal independent set. Otherwise choose a leaf v with neighbor u ; partitioning maximal sets by which of u, v they contain gives

$$\text{mis}(G) = \text{mis}(G - \{u, v\}) + \text{mis}(G - N[u]) \leq 2 \cdot 2^{(N-2)/2}.$$

For a general graph without isolates, choose a minimum-degree vertex v , of degree $\delta \geq 1$. Every maximal independent set meets $N[v]$; conditioning on a chosen member and overcounting gives

$$\text{mis}(G) \leq \sum_{u \in N[v]} \text{mis}(G - N[u]) \leq (\delta + 1) 3^{(N-\delta-1)/3} \leq 3^{N/3}.$$

The last inequality is $k \leq 3^{k/3}$ for integers $k \geq 2$, proved from $k = 2, 3, 4$ and the step $k + 3 \leq 3k$. These arguments are inductions on N .

The maximal members of any set family form an antichain. Counting permutations whose initial segment is a member of an antichain gives $\sum_{S \text{ maximal}} \binom{N}{|S|}^{-1} \leq 1$. Thus, writing $B_N = \binom{N}{\lfloor N/2 \rfloor}$,

$$S_{\text{fam}}(N) \leq \lceil 4(N+1)^4 \sqrt{B_N} \rceil, \quad (5)$$

and the same bound holds for complexes.

Graph lower bounds. For $q \geq 2$ and $t \geq 1$, let $G_{q,t}$ have a root with three neighbors w_1, w_2, w_3 . Attach t disjoint q -cliques to each w_i , joining w_i to exactly $q-1$ vertices of every attached clique. Then

$$|V(G_{q,t})| = 3qt + 4, \quad \alpha = 3t + 3, \quad I(G_{q,t}; x) = ((1+qx)^t + x(1+x)^t)^3 + x(1+qx)^{3t}.$$

For $q = 2$ this is Galvin's known tree $T_{3,t}$ [3]. Its top three coefficients, in descending order, are 1, A, B , where

$$A = 3(q^t + t), \quad B = q^{3t} + 3 \left(tq^{t-1} + \binom{t}{2} \right) + 3(q^t + t)^2.$$

After m isolates, the last log-concavity difference is

$$A^2 - B + Am + \frac{m(m+1)}{2}. \quad (6)$$

For fixed q , $B - A^2 \sim q^{3t}$ and $A = o(q^{3t/2})$. The exact positive root of (6) consequently gives the terminal threshold $\tau(G_{q,t}) \sim \sqrt{2} q^{3t/2}$. Since $\sigma \geq \tau$, the choices $q = 2, 3$ match (4) in exponential rate. To cover every N , choose $t = \lfloor (N-4)/(3q) \rfloor$.

An explicit finite lower bound is also available: for $q \geq 2, t \geq 8, A \leq 4q^t$ and $B - A^2 \geq 15q^{3t}/16$. At $m = \lfloor q^{3t/2} \rfloor$, we have $Am \leq q^{3t}/4$ and $m(m+1)/2 \leq 9q^{3t}/16$. Hence

$$S_{\text{for}}(N) > 2^{N/4-5/2} \quad (N \geq 52), \quad S_{\text{gr}}(N) > 3^{N/6-13/6} \quad (N \geq 76). \quad (7)$$

Complex lower bounds. For $N \geq 4$, put $r = \lfloor N/2 \rfloor$, $d = r + 2$, and fix a d -set $A \subseteq [N]$. The simplicial complex

$$\mathcal{K}_N = \{S \subseteq [N] : |S| \leq r\} \cup 2^A$$

has polynomial

$$P_{\mathcal{K}_N}(x) = \sum_{j=0}^r \binom{N}{j} x^j + dx^{d-1} + x^d.$$

Its top coefficients are $1, d, B_N$. By (6), its terminal threshold is $\sim \sqrt{2B_N} = 2^{N/2+o(N)}$, because $d = O(N)$ and $2^N/(N+1) \leq B_N \leq 2^N$. This proves both remaining lower bounds. For a finite version, $N \geq 16$ implies $B_N \geq 4d^2$: $2^N \geq 4(N+1)(N+2)^2$ follows by induction from $N = 16$. At $m = \lfloor \sqrt{B_N}/2 \rfloor$, the positive terms in (6) sum to at most $B_N/2$, while $B_N - d^2 \geq 3B_N/4$. Thus

$$S_{\text{cx}}(N) > \frac{1}{2} \sqrt{B_N} \geq \frac{2^{N/2}}{2\sqrt{N+1}} \quad (N \geq 16).$$

□

Sharpness in the maximal-member parameter. For the same graphs, put $x = q^t$. If the root is selected, there are x^3 maximal independent sets. If it is not selected, at least one w_i must be selected. In each branch, selecting w_i gives one maximal configuration; omitting it gives $x - 1$, since at least one selected clique vertex must dominate w_i . Thus

$$\text{mis}(G_{q,t}) = x^3 + \{x^3 - (x-1)^3\} = x^3 + 3x^2 - 3x + 1 \sim q^{3t}.$$

Therefore $\sigma(I(G_{q,t}; x)) \geq (\sqrt{2} + o(1)) \sqrt{\text{mis}(G_{q,t})}$. No bound $N^C M^{1/2-\varepsilon}$, with fixed C and $\varepsilon > 0$, can hold for all trees. The polynomial factor $4(N+1)^4$ in our upper bound is not claimed optimal.

6 Verification and limits of the claim

The accompanying exact-arithmetic verifier passed 191,941 checks, including all 138 families containing the empty set on one, two, or three elements. It checks the pairwise identity, its prescribed interior range, the baseline curvature, the endpoint constants, finite set-family counting bounds, and concrete global thresholds. These are checks supporting the written universal proof; they are not a proof-assistant certificate. In particular, a finite experiment is not used to infer either asymptotic theorem.

The result does not prove that the global and terminal thresholds agree for each graph. It proves that their *worst-case exponential rates* agree in the graph classes above. It does not resolve the number of log-concavity failures of an unmodified forest. The present argument and its audit were developed with ChatGPT and have not undergone independent mathematical review.

References

- [1] D. Handelman, *Spectral radii of primitive integral companion matrices and log concave polynomials*, Contemporary Mathematics 135 (1992), 231–237.
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- [4] C. Palmer and B. Patkós, *On the number of maximal independent sets: From Moon–Moser to Hujter–Tuza*, Journal of Graph Theory 102 (2023), 440–445; arXiv:2205.04082. <https://arxiv.org/abs/2205.04082>.